

ReactJS Hands-On - Props (Online Shopping App)

1. Objective

Create a React application named shoppingapp with a class component OnlineShopping that holds an array of 5 Cart items, and a class component Cart that receives Itemname and Price as props and displays them.

2. Prerequisites

Node.js, NPM, Visual Studio Code

3. Create the React Project

Command typed in the VS Code terminal:

```
npx create-react-app shoppingapp
```

4. Cart.js - Component Receiving Props

Cart is created under src/Components with two properties, Itemname and Price, passed in as props:

```
import React, { Component } from 'react';

class Cart extends Component {
  render() {
    return (
      <div>
        <p>Item Name: {this.props.Itemname}</p>
        <p>Price: {this.props.Price}</p>
      </div>
    );
  }
}

export default Cart;
```

5. OnlineShopping.js - Array of 5 Cart Items

OnlineShopping initializes an array of 5 items in its state and loops through them, passing each item's data to Cart as props:

```
import React, { Component } from 'react';
import Cart from './Cart';

class OnlineShopping extends Component {
  constructor(props) {
    super(props);
    this.state = {
      items: [
        { Itemname: 'Laptop', Price: 55000 },
        { Itemname: 'Mobile', Price: 15000 },
        { Itemname: 'Headphones', Price: 2000 },
        { Itemname: 'Keyboard', Price: 1200 },
      ]
    };
  }
}
```

```

        { Itemname: 'Mouse', Price: 800 }
      ];
    };
  }

  render() {
    return (
      <div>
        {this.state.items.map((item, index) => (
          <Cart key={index} Itemname={item.Itemname}
Price={item.Price} />
        ))}
      </div>
    );
  }
}

export default OnlineShopping;

```

6. App.js - Invoking OnlineShopping

```

import React from 'react';
import OnlineShopping from './Components/OnlineShopping';

function App() {
  return (
    <div className="App">
      <OnlineShopping />
    </div>
  );
}

export default App;

```

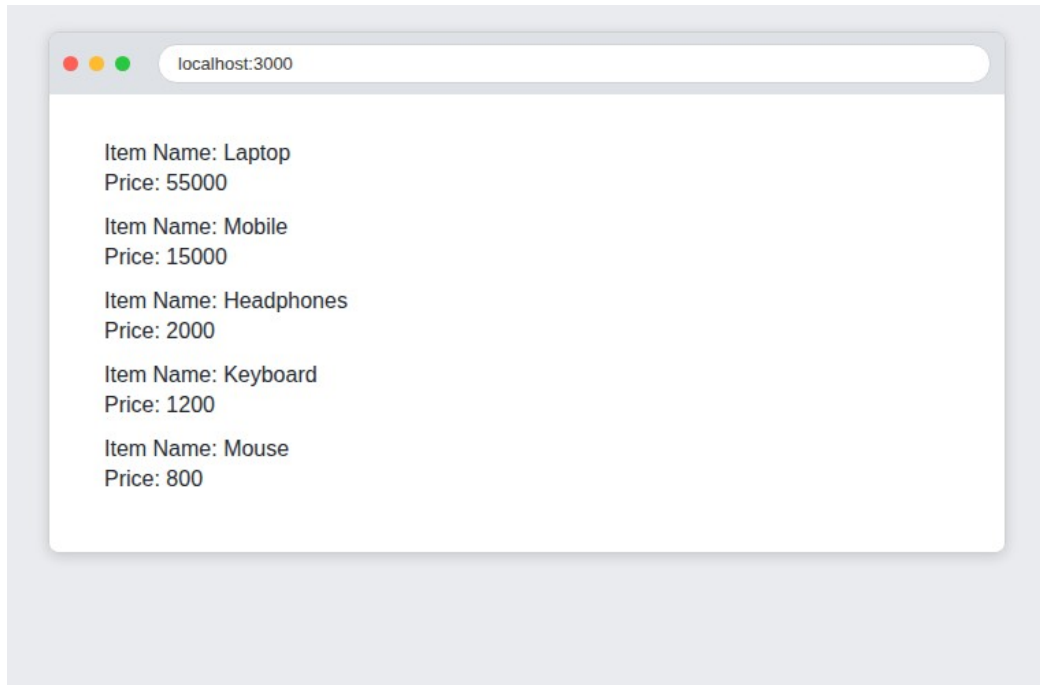
7. Run the Application

Command typed in the terminal, inside the shoppingapp folder:

```
npm start
```

8. Output - localhost:3000

Browser output after opening localhost:3000, showing all 5 cart items rendered through props:



9. Observation

The OnlineShopping component holds the array of 5 items in state and loops through it using `map()`, passing Itemname and Price to the Cart component as props on each iteration. Cart itself has no state of its own - it simply reads and displays the values received through `this.props`, which shows the difference between state (owned and managed by OnlineShopping) and props (read-only data passed down to Cart).